EECS 254 Final Project

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1. Problem description

In this assignment, I will employ unsupervised machine learning algorithms, to perform operations such as denoising, main frequency band analysis, feature extraction, and dimensionality reduction on electrocardiogram (ECG) data to distinguish different types of abnormal ECGs.

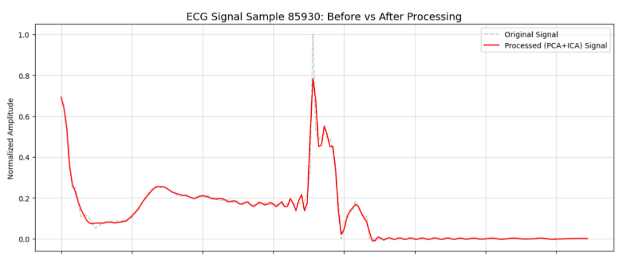
2. Dataset description

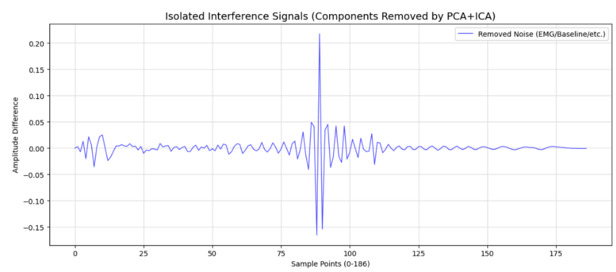
The **MIT-BIH Arrhythmia Database** is a collection of 48 half-hour excerpts of two-channel ambulatory ECG recordings, featuring over 110,000 manually annotated heartbeats categorized into five primary clinical classes (N, S, V, F, Q) to support the development and evaluation of automated arrhythmia detection algorithms. This dataset was collected from 47 subjects, with a sampling frequency of 360Hz.

3. Solution and results

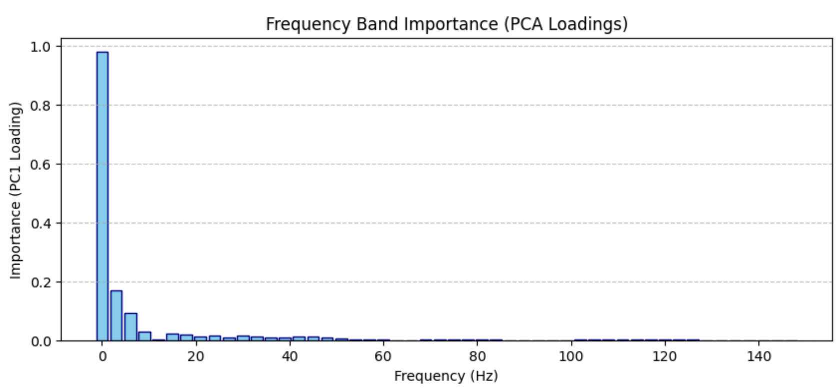
（1）**Denoising:** Use the **ICA algorithm** to remove the non-primary independent components in order to eliminate the interference from electromyography, electrode contact noise, baseline drift, etc. to the electrocardiogram.

The original signal X, the electrocardiogram signal after eliminating interference X1, and the interference signals are shown as the gray, red and blue lines in the figure below.



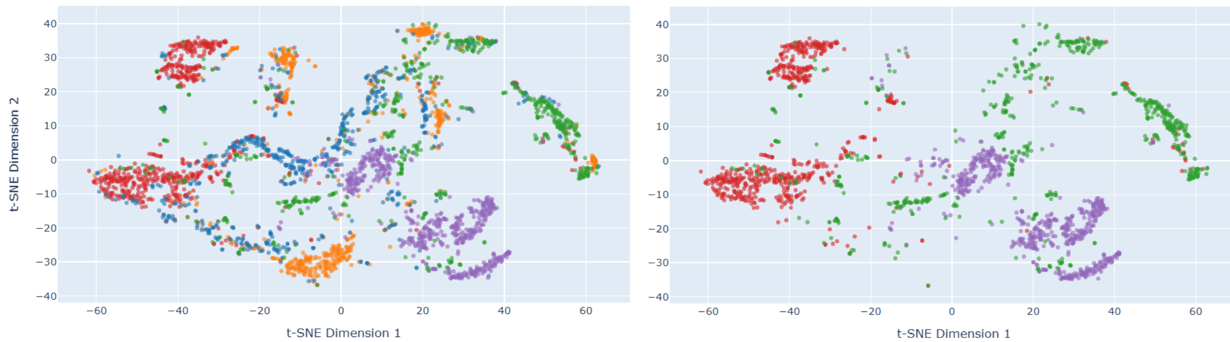


（2）**Frequency Principal Component Analysis:** The main frequency band of the ECG signal (0 to 150Hz) is downsampled at intervals of 3Hz, forming 50 band-pass filters with a bandwidth of 3Hz each. After passing through a band-pass filter, signal X1 generates 50 different frequency waveforms. The **PCA algorithm** is then applied to these waveforms to identify the frequency bands that contribute the most significant features. Extract the top 97% of the principal components and then apply filtering to X1 based on these components.

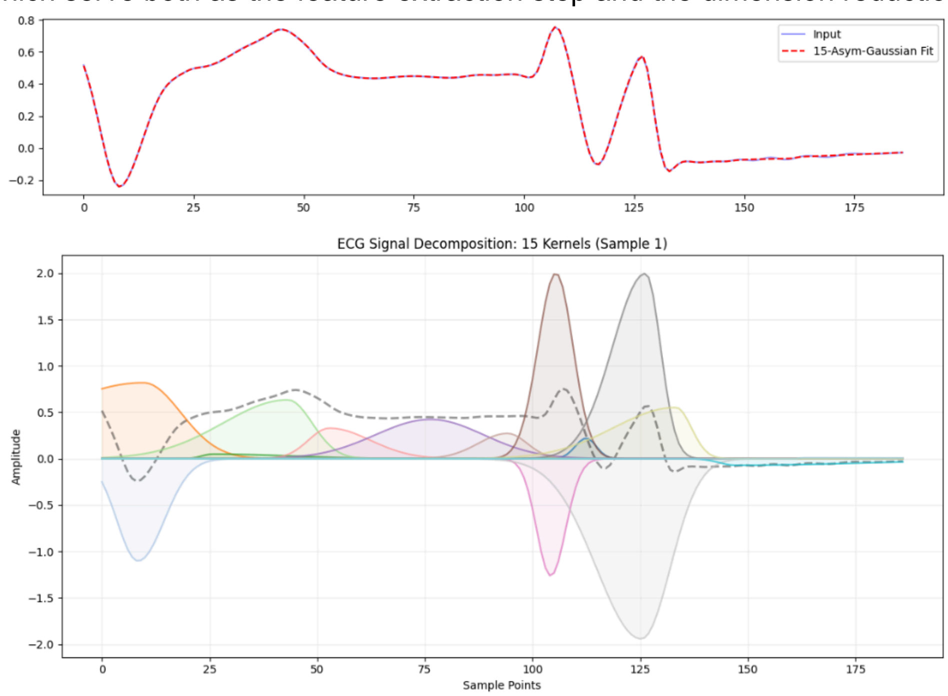


The results of the PCA analysis are shown in the figure above. It can be seen that the information of the electrocardiogram signal is concentrated in the low frequency range. If the **contribution rate is 97%**, then a low-pass filter ranging from **0 to 51 Hz** should be applied. If a 99% contribution level is adopted, a low-pass filter ranging from 0 to 120 Hz should be applied.

If the signal after low-pass filtering is directly processed using the **t-SNE algorithm**, the dimensionality is reduced from 187 to 2, as shown in the left figure below. The right side presents the annotation results of the three main abnormal electrocardiograms.



(3)**Feature extraction:** According to the original plan, 15 cases used **asymmetric Gaussian kernels** as **basis functions** to fit the 187-unit electrocardiogram waveform. Each nucleus contains 4 parameters: amplitude, position, left-side and right-side σ. Eventually, 60 values are obtained, which serve both as the feature extraction step and the dimension reduction step.

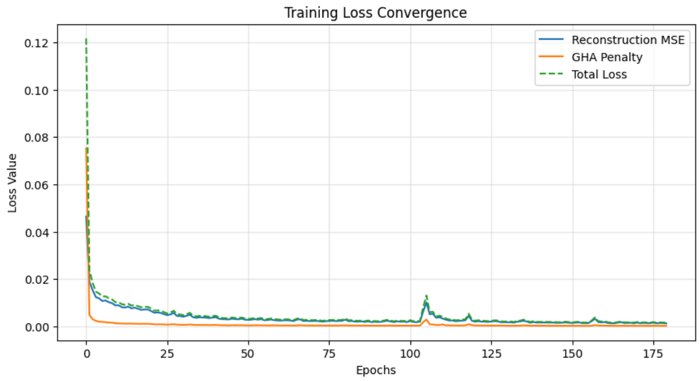


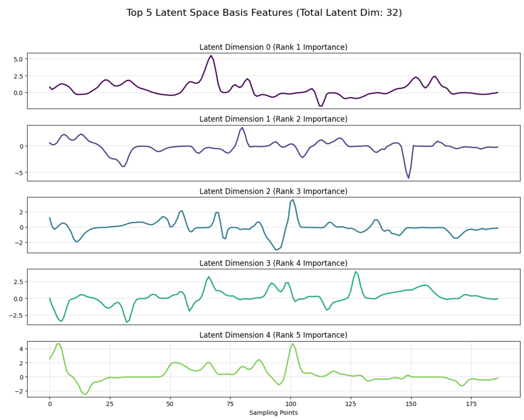
The fitting effect is shown in the figure along with 15 kernel functions. Although the fitting result is quite good, the effect of dimensionality reduction using these 60 feature values is very poor.
(4)**Feature extraction:** I decided to replace the basis functions with **convolutional autoencoders(CAE)** and combine it with the **GHA algorithm**. This method can also extract features and reduce dimensions simultaneously. However, its advantage lies in the ability to sort

the latent space features according to their contribution. The structure of the encoder and decoder is shown in the following table.

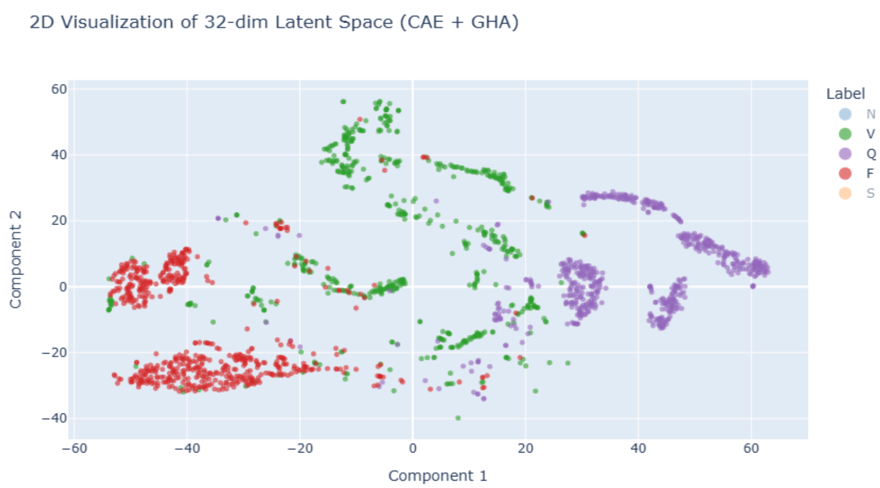
Layer	Encoder		Decoder	
	Operation	Output shape	Operation	Output shape
0	`input`	[3000,1,187]	`input`	[3000,32]
1	`nn.Conv1d(1, 16, kernel_size=11, stride=1, padding=5)`	[3000,16,187]	`nn.Linear(latent_dim, 256)`	[3000,256]
2	`nn.BatchNorm1d(16)`	[3000,16,187]	`nn.LeakyReLU(0.2)`	[3000,256]
3	`nn.LeakyReLU(0.2)`	[3000,16,187]	`nn.Linear(256, 64 * 46)`	[3000,2944]
4	`nn.MaxPool1d(2)`	[3000,16,93]	`nn.LeakyReLU(0.2)`	[3000,2944]
5	`nn.Conv1d(16, 32, kernel_size=7, stride=1, padding=3)`	[3000,32,93]	`nn.Unflatten(1, (64, 46))`	[3000,64,46]
6	`nn.BatchNorm1d(32)`	[3000,32,93]	`nn.Conv1d(64, 64, kernel_size=5, stride=1, padding=2)`	[3000,64,46]
7	`nn.LeakyReLU(0.2)`	[3000,32,93]	`nn.BatchNorm1d(64)`	[3000,64,46]
8	`nn.MaxPool1d(2)`	[3000,32,46]	`nn.LeakyReLU(0.2)`	[3000,64,46]
9	`nn.Conv1d(32, 64, kernel_size=5, stride=1, padding=2)`	[3000,64,46]	`nn.Upsample(size=93, mode='linear', align_corners=True)`	[3000,64,93]
10	`nn.BatchNorm1d(64)`	[3000,64,46]	`nn.Conv1d(64, 32, kernel_size=7, stride=1, padding=3)`	[3000,32,93]
11	`nn.LeakyReLU(0.2)`	[3000,64,46]	`nn.BatchNorm1d(32)`	[3000,32,93]
12	`nn.Flatten()`	[3000,2944]	`nn.LeakyReLU(0.2)`	[3000,32,93]
13	`nn.Linear(64 * 46, 256)`	[3000,256]	`n.Upsample(size=187, mode='linear', align_corners=True)`	[3000,32,187]
14	`nn.LeakyReLU(0.2)`	[3000,256]	`nn.Conv1d(32, 16, kernel_size=11, stride=1, padding=5)`	[3000,16,187]
15	`nn.Linear(256, 32)`	[3000,32]	`nn.BatchNorm1d(16)`	[3000,16,187]
16			`nn.LeakyReLU(0.2)`	[3000,16,187]
17			`nn.Conv1d(16, 1, kernel_size=3, stride=1, padding=1)`	[3000,1,187]
18			`nn.AdaptiveAvgPool1d(187)`	[3000,187]

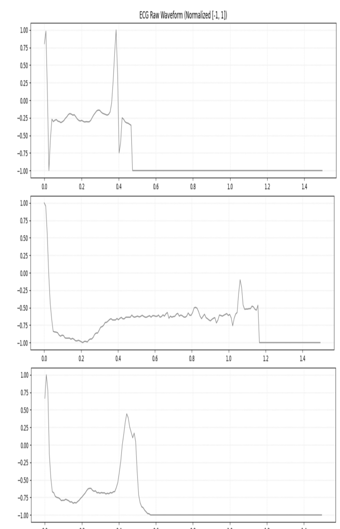
After the KL divergence constraint imposed by the GHA algorithm, the top five waveforms with the top 5 greatest contribution learned by the model are shown on the right side of the figure, and the decrease in the model's loss function is shown on the left side of the figure.





(5) **Dimensionality reduction:** The convolutional autoencoder reduces the dimension from 187 to 32. Then, the **t-SNE algorithm** is used to further reduce the dimension from 32 to 2.





For the **three main abnormal heartbeats**, the annotation results at this time and waveforms are as shown in the above figure. The three heartbeats are respectively located in **the left, middle and right areas**, with clear boundary. It can be seen that this unsupervised algorithm process can clearly distinguish different types of abnormal heartbeats.

By directly applying t-SNE to the waveform and to the features obtained after applying CAE + GHA, the **Calinski-Harabasz Index** increased **from 443.35 to 527.14**. The ratio of the inter-class distance to the intra-class distance increased, indicating the effectiveness of the feature extraction and dimensionality reduction algorithms. The initial problem was resolved.

The reproducible code can be obtained at the following link or another PDF:
https://colab.research.google.com/drive/1BLQUnmtg_IgF5Yvcggb4mdmKeNYVLGNz?usp=drive_link